Data Mining Project

Context

French public administrations publish open datasets that can be used for machine learning tasks. Your objective is to select one dataset from the Données pour l'apprentissage automatique collection on data.gouv.fr, explore it, prepare it, and build a predictive or descriptive model.

You will work as a data mining practitioner dealing with real, imperfect, heterogeneous data.

Dataset Selection

Choose one dataset from the “Données pour l’apprentissage automatique” collection.

Each group must select a unique thematic (no thematic may be used by more than one group).

Project Statement

1. Dataset Selection and Problem Definition

Choose a dataset and define a clear data mining problem. Examples:

- Predict air pollution levels using meteorological variables.
- Classify municipalities based on socio-economic indicators.
- Predict building energy consumption.
- Cluster transportation routes based on duration or frequency.

You must specify:

- A precise question
- The modelling objective
- The target variables
- The explanatory variables

2. Exploratory Data Analysis

Expected deliverables:

- Descriptive statistics
- Relevant visualizations (distributions, correlations, boxplots, etc.)
- Analysis of missing values
- Identification of anomalies or biases
- Selection of relevant features

3. Data Preparation

Include:

- Cleaning (missing values, duplicates, outliers)
- Encoding of categorical variables
- Normalization or standardization
- Feature engineering when appropriate
- Train/test split

4. Modelling

Depending on your problem:

- **Classification**
- **Regression**
- **Clustering**

Expected outputs:

- Comparison of several models
- Selection of the final model
- Evaluation metrics

5. Interpretation and Communication

You must produce:

- Interpretation of results
- Discussion of model limitations
- Recommendations for future improvements
- A 10-minute oral presentation of your work

Final Deliverables

1. A 10 pages max summary report
2. A short oral presentation (slides in pdf format)